

PROBLEM SET 2—SOLUTIONS

ECON301—ECONOMETRICS I

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Instructions: Due on 25/02/2020 before 12 PM in the envelope pasted to my office door. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission. Don't forget to attach the cover sheet.

1. Suppose you have the following data on 11 student's combined SAT scores X and their cumulative grade point average at graduation Y .¹

Y_i	X_i
3.63	1490
2.37	1300
3.33	1510
3.32	1420
3.27	1490
2.37	1180
3.61	1550
3.23	1460
2.59	1300
3.30	1450
3.21	1550

- (a) Write down the econometric model relating X and Y (i.e. the linear regression model).
(b) Calculate the following:

$$n; \bar{X}; \bar{Y}; \sum (X_i - \bar{X})(Y_i - \bar{Y}); \sum (X_i - \bar{X})^2; \hat{\beta}_0 \text{ and } \hat{\beta}_1.$$

- (c) Calculate $\hat{Y}_i$ and $\hat{u}_i$ for each observation.
(d) Calculate R^2 .
(e) Compute $\sum \hat{u}_i$. Explain why do you get this result.

Solution

- (a) The linear regression model relating X and Y is

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

- (b) It is useful to construct a table:

¹You can use any spreadsheet software to calculate what is asked in this exercise.

	Y_i	X_i	$(X_i - \bar{X})$	$(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$	$(X_i - \bar{X})(Y_i - \bar{Y})$
1	3.62	1490	63	0.51	3935	31.9
2	2.37	1300	-127	-0.74	16198	94.3
3	3.33	1510	83	0.22	6844	18.1
4	3.32	1420	-7	0.21	53	-1.5
5	3.27	1490	63	0.16	3935	10.0
6	2.37	1180	-247	-0.74	61144	183.2
7	3.61	1550	123	0.50	15062	61.3
8	3.23	1460	33	0.12	1071	3.9
9	2.59	1300	-127	-0.52	16198	66.3
10	3.3	1450	23	0.19	517	4.3
11	3.21	1550	123	0.10	15062	12.2
MEAN	3.11	1427	0	0	12729	44
$\sum_{i=1}^{11}$	34.22	15700	0	0	140018	484

	$\hat{Y}_i$	$\hat{u}_i$	$(Y_i - \bar{Y})^2$	$\hat{Y}_i - \bar{Y}$	$(\hat{Y}_i - \bar{Y})^2$	$\hat{u}_i^2$
1	3.33	0.29	0.26	0.22	0.047	0.085
2	2.67	-0.30	0.55	-0.44	0.193	0.091
3	3.40	-0.07	0.05	0.29	0.082	0.004
4	3.09	0.23	0.04	-0.03	0.001	0.055
5	3.33	-0.06	0.03	0.22	0.047	0.003
6	2.26	0.11	0.55	-0.85	0.730	0.013
7	3.54	0.07	0.25	0.42	0.180	0.006
8	3.22	0.01	0.01	0.11	0.013	0.000
9	2.67	-0.08	0.27	-0.44	0.193	0.007
10	3.19	0.11	0.04	0.08	0.006	0.012
11	3.54	-0.33	0.01	0.42	0.180	0.106
MEAN	3	0	0.19	0	0.152	0.035
$\sum_{i=1}^{11}$	34	0	2.054	0	1.673	0.382

Then looking at the tables above we have:

- $n = 11$
- $\bar{X} = 1427$
- $\bar{Y} = 3.11$
- $\sum (X_i - \bar{X})(Y_i - \bar{Y}) = 484$
- $\sum (X_i - \bar{X})^2 = 140018$
- $\hat{\beta}_0 = -1.82$ and $\hat{\beta}_1 = 0.003$

(c) See table above.

(d) From the second table above we know

$$TSS = 2.054, \quad ESS = 1.673, \quad SSR = 0.382.$$

Therefore using the formula for R^2 we get

$$R^2 = \frac{ESS}{TSS} = 0.81,$$

alternatively,

$$R^2 = 1 - \frac{SSR}{TSS} = 0.81.$$

(e) $\sum_{i=1}^{11} \hat{u}_i = 0$ (See the second table above).

This is always true when we run an OLS regression. In other words, this is an implication of the use of OLS to compute estimators. Since OLS seeks to minimize the distance between the observations and the regression line, what OLS is actually doing is minimizing the "mistakes" that our estimated model produces so that on average this mistake is zero or equivalently:

$$\frac{\sum_{i=1}^n \hat{u}_i}{n} = \frac{0}{n}$$

A more formal argument is as follows:

$$\begin{aligned}\sum_{i=1}^n \hat{u}_i &= \sum_{i=1}^n (Y_i - \hat{Y}_i) \\ &= \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) \\ &= \sum_{i=1}^n (Y_i - \bar{Y} + \hat{\beta}_1 \bar{X} - \hat{\beta}_1 X_i) \\ &= \sum_{i=1}^n Y_i - \sum_{i=1}^n \bar{Y} + \sum_{i=1}^n \hat{\beta}_1 \bar{X} - \sum_{i=1}^n \hat{\beta}_1 X_i \\ &= n\bar{Y} - n\bar{Y} + n\hat{\beta}_1 \bar{X} - n\hat{\beta}_1 \bar{X} \\ &= 0\end{aligned}$$

Where we have used:

$$\frac{1}{n} \sum_{i=1}^n X_i = \bar{X} \Rightarrow \sum_{i=1}^n X_i = n\bar{X}$$

$$\frac{1}{n} \sum_{i=1}^n Y_i = \bar{Y} \Rightarrow \sum_{i=1}^n Y_i = n\bar{Y}$$

$$\sum_{i=1}^n c = nc,$$

for any constant c .

2. Do problems 4, 6 and 12 from Chapter 2 in the book.

Solution

Problem 4. (i) When $cigs = 0$, predicted birth weight is 119.97 ounces. When $cigs = 20$, $\widehat{bwght} = 109.49$. This is about an 8.6% drop.

(ii) Not necessarily. Formally, is difficult to argue that $E(u|cigs) = 0$. There are many other factors that can affect birth weight, particularly overall health of the mother and quality of prenatal care. These could be correlated with cigarette smoking during birth. Also, something such as caffeine consumption can affect birth weight, and might also be correlated with cigarette smoking.

(iii) If we want a predicted $bwght$ of 125, then $cigs = (125 - 119.77)/(-.524) \approx -10.18$, or about -10 cigarettes. This is nonsense, of course, and it shows what happens when we are trying to predict something as complicated as birth weight with only a single explanatory variable. The largest predicted birth weight is necessarily 119.77. Actually, if you check the data (which is part of the `wooldridge` package in R), you will be able to see that almost 700 of the births in the sample had a birth weight higher than 119.97.

(iv) If you check the data, you will see that 1,176 out of 1,388 women did not smoke while pregnant, or about 84.7%. Because we are using only $cigs$ to explain birth weight, we have only one predicted birth weight at $cigs = 0$. The predicted birth weight is necessarily roughly in the middle of the observed birth weights at $cigs = 0$, and so we will under predict high birth weights.

Problem 6. (i) Yes. If living closer to an incinerator depresses housing prices, then being farther away increases housing prices. More specifically, the coefficient on $\log(dist)$ is an elasticity: for every 1% increase in distance, the price increases by 0.312%.

(ii) Formally, what you are being asked to think about is whether assumption SLR4 holds: $E(u|\log(dist)) = 0$. If the city chooses to locate the incinerator in an area away from more expensive neighborhoods, then $\log(dist)$ is positively correlated with housing quality. This would violate SLR.4, and OLS estimation is biased, and furthermore, we are not distilling the ceteris paribus or causal effect of distance on price.

(iii) Size of the house, number of bathrooms, size of the lot, age of the home, and quality of the neighborhood (including school quality), are just a handful of factors. As mentioned in part (ii), these could certainly be correlated with $dist$ and, of course, with $\log(dist)$.

Problem 12. (i) Define for every b_0 , $Q(b_0) = \sum_{i=1}^n (y_i - b_0)^2$. Then the problem is

$$\min_{b_0} Q(b_0) \tag{1}$$

Let $\tilde{\beta}_0 \in \arg \min_{b_0} Q(b_0)$. We will show that $\tilde{\beta}_0 = \bar{Y}$. A necessary condition for $\tilde{\beta}_0$ to solve problem (1) is that the derivative of $Q(b_0)$ with respect to b_0 must be zero when evaluated at $\tilde{\beta}_0$, i.e., the first order condition is:

$$\frac{dQ(\tilde{\beta}_0)}{db_0} = 0.$$

Using the chain rule to calculate the derivative, the above equation becomes:

$$-2 \sum_{i=1}^n (y_i - \tilde{\beta}_0) = 0$$

Now divide the equation by $2n$ to get

$$\begin{aligned}\frac{1}{n} \sum_{i=1}^n (y_i - \tilde{\beta}_0) &= 0 \\ \bar{Y} - \frac{n\tilde{\beta}_0}{n} &= 0 \\ \bar{Y} &= \tilde{\beta}_0.\end{aligned}$$

The first order condition is only necessary but not sufficient. To verify that $\tilde{\beta}_0 = \bar{Y}$ indeed achieves a minimum, we will use the second-order criterion for a minimization problem, i.e., we will check that the second derivative of the objective function, when evaluated at $\tilde{\beta}_0$, has a positive sign:

$$\frac{d^2 Q(\tilde{\beta}_0)}{d\tilde{\beta}_0^2} = 2n > 0$$

Since the derivative is positive^a at $\tilde{\beta}_0$, then indeed it is a minimum. This completes the proof.

(ii) Define, for each i , $\tilde{u}_i = y_i - \bar{Y}$. We need to show that $\sum_{i=1}^n \tilde{u}_i = 0$.

$$\sum_{i=1}^n \tilde{u}_i = \sum_{i=1}^n (y_i - \bar{Y}) = \sum_{i=1}^n y_i - n\bar{Y} = n\bar{Y} - n\bar{Y} = 0.$$

^aActually, the objective function is convex at every point.

3. Suppose that you have a random sample $\{(x_i, y_i) : i = 1, \dots, n\}$. Consider the following problem:

$$\min_{b_0, b_1} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2 \quad (2)$$

where b_0 and b_1 are dummy arguments for the optimization problems.

Show that the OLS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are the solution to problem (2). Be as rigorous as you can in your proof, i.e., you will have to show that indeed $(\hat{\beta}_0, \hat{\beta}_1)$ achieves a minimum in the objective of problem (2).

Solution

Denote the objective in (2) as $Q(b_0, b_1)$. The goal is then to solve the problem $\min_{b_0, b_1} Q(b_0, b_1)$. From a fundamental result from multivariate calculus we know that a necessary condition for $\hat{\beta}_0$ and $\hat{\beta}_1$ to solve the minimization problems is that the partial derivatives of $Q(b_0, b_1)$ with respect to b_0 and b_1 must be zero when evaluated at $\hat{\beta}_0, \hat{\beta}_1$, i.e., the first order conditions:

$$\begin{aligned}\frac{\partial Q(\hat{\beta}_0, \hat{\beta}_1)}{\partial b_0} &= 0 \\ \frac{\partial Q(\hat{\beta}_0, \hat{\beta}_1)}{\partial b_1} &= 0\end{aligned}$$

Using the chain rule to calculate both partial derivatives, the two equations above become

$$-2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \quad (3)$$

$$-2 \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0. \quad (4)$$

If we divide by $2n$ both equations, we get exactly the same conditions that we obtained when using the

method of moments. Thus, as you can verify, the solution to the system of equations (3) is the OLS estimator:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}, \quad \hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{X})^2}$$

The first order conditions are only necessary but not sufficient conditions. We can verify that $\hat{\beta}_0$ and $\hat{\beta}_1$ indeed have achieved a minimum in problem (2) by noticing that, for any b_0 and b_1 :

$$\begin{aligned} Q(b_0, b_1) &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i + (\hat{\beta}_0 - b_0) + (\hat{\beta}_1 - b_1)x_i)^2 \\ &= \sum_{i=1}^n (\hat{u}_i + (\hat{\beta}_0 - b_0) + (\hat{\beta}_1 - b_1)x_i)^2 \\ &= \sum_{i=1}^n \hat{u}_i^2 + n(\hat{\beta}_0 - b_0)^2 + (\hat{\beta}_1 - b_1) \sum_{i=1}^n x_i^2 + 2(\hat{\beta}_0 - b_0)(\hat{\beta}_1 - b_1) \sum_{i=1}^n x_i, \end{aligned}$$

where I have used the fact that $\sum_{i=1}^n \hat{u}_i = 0$ and $\sum_{i=1}^n x_i \hat{u}_i = 0$. The first term does not depend on b_0 or b_1 whereas the sum of the last three terms can be written as

$$\sum_{i=1}^n ((\hat{\beta}_0 - b_0) + (\hat{\beta}_1 - b_1)x_i)^2,$$

as you can easily verify using algebra. Since this term is a sum of squares, the smallest it can be is zero. Hence, it will be at its smallest value when $b_0 = \hat{\beta}_0$ and $b_1 = \hat{\beta}_1$.

4. Suppose that a researcher, using data on class size (CS) and average test scores from 100 third-grade classes, estimates the OLS Regression

$$\widehat{TestScore} = 520.4 - \frac{5.82}{(20.4)} CS, \quad R^2 = 0.08, RSE = 11.5,$$

where the standard errors for the coefficient estimates are given by the numbers in parentheses.

- Suppose a classroom has 20 students. What is the regression's prediction for that classroom's average test score?
- Last year a classroom had 19 students, and this year it has 21 students. What is the regression's prediction for the change in the classroom average test score?
- The sample average class size across the 100 classrooms is 21.4. What is the sample average of the test scores across the 100 classrooms? (*Hint*: Review the formulas for the OLS estimators).
- What is the sample standard deviation of the test scores across the 100 classrooms? (*Hint*: Review the formulas for the R^2 and RSE).

Solution

- (a) The regression's prediction for that classroom test score is

$$\widehat{TestScore} = 520.4 - 5.82 \times (20) = 404.$$

- (b) The prediction for last year was

$$\widehat{TestScore} = 520.4 - 5.82 \times (19) = 409.82.$$

while the prediction for this year is

$$\widehat{TestScore} = 520.4 - 5.82 \times (21) = 398.18.$$

Thus, the predicted change in the average test score in the classroom is

$$\Delta \widehat{TestScore} = 398.18 - 409.82 = -11.64.$$

Alternatively, using $\widehat{\Delta TestScores} = \hat{\beta}_1 \times \widehat{\Delta CS}$ we get

$$\widehat{\Delta TestScores} = -5.82 \times 2 = -11.64.$$

(c) From the formula for $\hat{\beta}_0$ we have

$$\overline{TestScores} = \hat{\beta}_0 - \hat{\beta}_1 \times \overline{CS} = 520.4 - 5.82 \times 21.4 = 395.852.$$

(d) We can use the formula for the RSE to calculate the sum of squared residuals SSR

$$SSR = (n - 1)RSE^2 = 98 \times 11.5^2 = 12960.5$$

We can use the formula for R^2 to compute the total sum of squares TSS

$$TSS = \frac{SSR}{1 - R^2} = \frac{12960.5}{1 - 0.08} = \frac{12960.5}{0.92} = 14086.95.$$

Thus, the sample variance and standard deviation are

$$s^2 = \frac{TSS}{n - 1} = \frac{14086.95}{99} = 142.29, \quad s = \sqrt{142.29} = 11.92.$$

5. **Optional.** Do the empirical exercise C9 from chapter 2 in the book. You will have to attach a printout of your R code and output.

(i) Below is the necessary code to answer the questions

```
rm(list=ls())
library(wooldridge)
library(tidyverse)

data("countymurders")

num0murder<-countymurders%>%filter(murders==0)%>%nrow()
numatleast1exec<-countymurders%>%filter(execs>=1)%>%nrow()
summary(select(countymurders,execs))
```

There are 15651 counties had zero murders in 1996. There are 205 counties had at least one execution. The largest number of executions is 7.

(ii) The R code to estimate the model is

```
regression5<-lm(murders~execs,data=countymurders)
summary(regression5)
```

The estimated equation is

$$\widehat{murders} = \underset{(0.2418)}{6.8382} - \underset{(2.1463)}{65.4650}execs, \quad R^2 = 0.02431, RSE = 46.64, n = 37349$$

(iii) The coefficient on *execs* is 65.46, which is positive. It means that one more execution in one county will have 65.46 more murders. Thus, it doesn't suggest that capital punishment is a deterrent for murder.

(iv) When *execs* = 0, the smallest number of murders that can be predicted by the equation is 6.84. The residual for a county with zero executions and zero murders is -6.8382.

(v) In this single variable regression, it is highly suspected that the error term is correlated with the factor 'execution', which violates the assumption SLR4 for OLS, i.e. $E(u|execs) \neq 0$. For example, in a poor county it is much likely that it has both a large number of executions and a large number of murders. In order to isolate this correlation, we need add some other control factors, which further need a multiple regression.